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REVIEW

Examing the Anticancer Potential of Punica Granatum Linn: a Review

M. Padmaa Paarakh[1]*, Preethy Ani Jose[2]

[1] Department of Pharmacognosy, The Oxford College of Pharmacy, Karnataka, India.

[2] Department of Pharmaceutics, The Oxford College of Pharmacy, Karnataka, India.

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Corresponding Author : M. Padmaa Paarakh

Email : ardhendu_mandal@yahoo.co.in

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Abstract:

Medicinal plants are used in traditional medical systems and are considered to be the source of healthy human life. *Punica granatum* Linn is one of the potential medicinal plants that is not only consumed as a fruit but also used to treat diseases. The parts of the plant used in the activity are root bark, flower buds, fruit, and fruit skin. A large number of phytoconstituents are isolated from plants, namely punicalin, punicaligin, ellagic acid, granatin B, galagyl dilactone,casuarin, pedunculagin, gulantin A, tellimagrandin I, gallic acid, corilagin and pentunidin. The pharmacological activities reported so far include antioxidant effects, anticarcinogenic effects, anti-inflammatory effects, cardiovascular diseases, diabetes, dental diseases, anthelmintic effects, antireproductive effects, gastroprotective effects, antifungal effects, analgesic effects, these include hypoglycemic and arteriosclerotic effects. The purpose of this study is to provide an overview of *Punica granatum* in relation to the anticancer activities reported so far.

Keywords : *Punica granatum*, Anti-cancer activity, phytoconstituents

Introduction :

Cancer remains the leading cause of death worldwide, and only modest progress has been made in reducing morbidity and mortality from this disease. Cancer is caused by three factors: improper diet, genetics, and environmental factors. Up to 95% of all cancers are caused by lifestyle and can take up to 20-30 years to develop. According to the latest estimates from the American Cancer Society and the International Union Against Cancer, 12 million people were diagnosed with cancer and 7 million died from cancer worldwide last year. These numbers are expected to double by 2030 (27 million infections, 17 million deaths). According to a World Health Organization report, more than 80% of the world's population relies on traditional medicine for primary health care [1].

Natural drugs that are part of the fight against cancer include vinca alkaloids (vincristine, vinblastine, vindesine, vinorelbine), taxanes (paclitaxel, docetaxel), podophyllotoxin and its derivatives (etoposide, teniposide), camptothecin and Includes its derivatives (topotecan, irinotecan), anthracyclines (doxorubicin, daunorubicin, epirubicin, idarubicin), etc.

Extensive research is being conducted to treat cancer, and much of the research has been successful in treating patients. Side effects and drug resistance are the biggest threats to healthcare professionals working on cancer treatment, and there is a great need to find new anti-cancer drugs to alleviate patient pain and suffering.

This review focuses on the traditional uses of *Punica granatum* and the botanical components isolated so far, as well as a detailed study of its anticancer effects. *Punica granatum* Linn (Pomegranate family), commonly known as pomegranate, is a large deciduous shrub or small tree, 5 to 10 m in height, found wild and cultivated in hilly areas up to 2000 m above sea level throughout India [2].

P. granatum contains many chemical components that interact in complex ways to produce pharmacological activity. Many active ingredients that confer medicinal properties have been isolated and characterized.

Active ingredients isolated from different parts of *Punica granatum*:

Fruit: Consists of a rich variety of flavonoids, accounting for approximately 0.2% to 1.0% of the fruit. Approximately 30% of the total anthocyanidins contained in pomegranates are contained in the skin. Nicotinic acid, pectin, protein, riboflavin, thiamine, vitamin C, delphinidin, aspartic acid, citric acid, ellagic acid, gallic acid, malic acid, glutamine, isoquercetin.

Seeds: Isoflavones such as genistein, diazine, diazein, genistin, and metabolic derivatives of estradiol, estrone and punicic acid.

Stems and roots: alkaloids such as isopelletierin, pseudopellethierin, N-methylisopelltierin, anthocyanidins, pelargonidin, elagotannin, gallic acid, ellagic acid.

Flower: Asiatic acid, conic acid, pelargonidin 3,5-diglucoside, sitosterol and its β-D-glycosides.

Bark: betulinic acid, galtan-3-one, isopellecherine, pseudopellecherine, methyl isopellecherine, perecherine, friedelin.

Leaf: granatin A and B and punicatrin. Fruit skin: punicalin, punicaligin, ellagic acid, garnetin B, galagyl dilactone, casuarine, pedunculagin, gulantin A, tellimaglandin I, gallic acid, corilagin, pentunidin. There have also been some systematic scientific studies on the effectiveness of the whole thing. The plant or its parts are extracted in the form of various extracts and used to treat various diseases. This plant has antibacterial, antifungal, hypoglycemic, antioxidant, hypolipidemic, analgesic, immunomodulatory, antispasmodic, anthelmintic, antireproductive, antidiabetic, antiinflammatory, gastric It has been reported to have protective, uterostimulatory, cytotoxic, carcinogenic, antiangiogenic, antiatherosclerotic, and antihypertensive effects.

Carotid artery stenosis, myocardial perfusion, dental disease, ultraviolet light, erectile dysfunction, male fertility, neonatal hypoxia, ischemic brain injury, Alzheimer's disease, obesity, anticancer activity. Review of different methods of green synthesis using nanoparticles. Punica granatum and Muraya koenigii Botanical Extracts and Their Applications also describes the uses of this plant [3].

Detailed overview of the anticancer activity of *Punica granatum*:

Many modern anticancer drugs are analogs of natural compounds. Proper combination of drugs and natural ingredients can improve the overall effectiveness of treatment. It is believed that plant medicines rich in polyphenols and flavonoid molecules may help protect normal cells from the harmful effects of anticancer drugs. In vitro studies using three prostate cancer cell lines (DU-145, LNCaP, and PC-3) showed that various pomegranate extracts (juice, seed oil, and peel) inhibited prostate cancer cell invasion and proliferation, in other words, it was shown to effectively inhibit cell destruction. This cycle triggers and induces apoptosis and inhibits tumor growth. These studies also showed that a combination of pomegranate extracts from different parts of the fruit was more effective than a single extract [4].

Pomegranate polyphenol [Pg], ellagitannin-rich extract, and whole fruit juice extract inhibited the gene expression of HSD3B2 (3β-hydroxysteroid dehydrogenase). These are important androgen synthases in human prostate cancer cells LNCaP, LNCaP-AR, and DU-145. Pg may have anticarcinogenic effects because it inhibits CYP activity/expression required for activation of carcinogens [5].

Treatment with (50-150 μg/ml) pomegranate fruit extract (PFE) for 72 hours' dose-dependently arrested cells in the G0/G1 phase of the cell cycle and significantly inhibited lung cancer. Decreased protein expression of WAF1/p21 and KIP1/p27, decreased protein expression of cyclin D1, D2 and E, decreased expression of cyclin-dependent kinase (cdk) 2, cdk4 and cdk6, phosphorylation of MAPK proteins, decreased protein expression of PI3K. Inhibition of Akt phosphorylation of Thr308, NF-kB, and IKK alpha, IκB, Ki-67, and PCNA degradation and phosphorylation [2].Several animal studies have elucidated the potential anticancer mechanisms of pomegranate. Two studies using mice implanted with the prostate cancer cell line PC-3 showed that pomegranate fruit extract (PFE; the edible part of the fruit, excluding the pericarp) stimulates cells by modulating proteins that regulate apoptosis was shown to inhibit the proliferation of and induce apoptosis [6].

In vitro studies have shown that multiple PFEs inhibit prostate cancer cell proliferation, induce apoptosis in several prostate cancer cell lines (including the highly aggressive PC-3 prostate cancer cells), and Prostate cancer cells are reduced by suppressing the invasive ability of PC-3 cells and proliferation of DU-145.

When equal amounts of fermented pomegranate juice (FPJ), pomegranate peel extract (PPE), and cold-pressed seed oil are combined, the extract (CPSO) undergoes DU- 145 Invasion of prostate cancer cells was inhibited by 99%. CPSO extract or FPJ extract alone inhibited infiltration by 60%, and the combination of any two extracts resulted in 90% inhibition. Studies in mice have also shown that PFE inhibits prostate tumor growth and reduces PSA levels [7].

In an open-label phase II clinical trial of 46 men with recurrent prostate cancer, 16 patients (35%) had lower serum prostate levels during treatment of pomegranate juice. Specific antigen (PSA) levels were significantly reduced (mean = 27%). Corresponding in vitro studies using patient plasma and serum showed significantly reduced proliferation and increased apoptosis in prostate cancer cell lines. Nitric oxide intake through pomegranate polyphenol intake was significantly correlated with lower PSA levels. These results suggest that pomegranates may be effective against prostate cancer due to their anti-proliferative, apoptotic, antioxidant, and possibly anti-inflammatory effects [3].

In vitro studies using patient serum investigated whether pomegranate juice [PJ] ingestion affected the proliferation rate or apoptosis of LNCaP prostate cancer cells in culture. Serum taken 9 months into the study and incubated with LNCaP reduced cell proliferation by an average of 12% compared to baseline in 84% of patients. Additionally, an average 17.5% increase in apoptosis was observed in her 75% of patients. This study suggests that PJ or PJ components may be a promising treatment for prostate cancer, especially recurrent forms of prostate cancer where PSA levels are elevated [8].

Pomegranate components were studied to inhibit angiogenesis by downregulating vascular endothelial growth factor in MCF7 breast cancer and human umbilical vein endothelial cell lines.

The anti-angiogenic potential of Pg was enhanced by vascular endothelial growth factor (VEGF), IL-4 and migration inhibitory factor (MIF) -231) Human breast cancer cells and immortalized normal human breast epithelial cells (MCF-10A). VEGF was significantly reduced in his MCF-10A and MCF-7.

However, MIF was increased in MDA-MB-231, demonstrating the remarkable potential of the inhibitory effect of angiogenesis by Pg fraction on human umbilical vein endothelial cells (HUVECs) [7]. In HT-29 colon cancer cells, activation of nuclear factor kappa B (NFκB) by tumor necrosis factor alpha (TNFα) increases the expression of cyclooxygenase 2 (COX-2). This is the inflammatory cell signaling process that may be responsible. The state of cancer development and progression. Treatment of HT-29 colon cancer cells with PJ, total pomegranate tannins, or concentrated pomegranate punicalagin significantly reduced COX-2 expression. PJ treatment resulted in the highest COX-2-inhibition (79%) compared to single-component treatment. This effect is thought to be due to the synergistic activity of bioactive components that are thought to be necessary for the anti-inflammatory and anti-carcinogenic effects of pomegranate [9].

Another in vitro study investigated the effects of punicalagin, ellagic acid, total pomegranate tannins, and PJ on several cell lines. All formulations reduced the viable cell numbers of oral cancer cell lines KB and CAL-27 and colon cancer cell lines HT-29 and HCT-116, but a higher degree of inhibition was achieved with pure PJ. The synergistic effect of its physiologically active ingredients. Studies using breast cancer cell lines MCF-7 and MB-MDA-231 showed that pomegranate components effectively inhibited angiogenesis, tumor growth, proliferation and invasiveness has been shown to induce apoptosis.

To investigate the effects of FPJ and CPSO extracts and HPLC-separated peak B (from fruit extract), mouse mammary organ cultures, an animal model of breast cancer, were used to investigate the effects of Prediction accuracy was over 75%. In vivo carcinogenicity was investigated [2]. They found that cancer glands treated with each pomegranate compound had lower lesion incidence, ranging from 37 percent at FPJ to 75 to 90 percent at Peak B and CPSO.

The seed oil is primarily composed of punicic acid, a trienoic acid with anticarcinogenic properties that are effective at very low doses (1 mg/ml in organ cultures). Peak B is considered to be a phenolic compound with potent chemopreventive properties [10].

Studies in mice have shown that PFE inhibits tumor formation in lung and skin cancer models. In a lung cancer study, mice that were given daily oral doses of PFE that could reasonably be consumed by humans (exact doses are not available) had less lung tumor growth than mice that did not receive PFE was shown to be significantly less [4].

In mice treated with 12-otetradecanoylphorbol-13-acetate (TPA), which causes skin cancer, the incidence of skin tumors was lower in mice treated topically with PFE. It was significantly lower. In the PFE treatment group, only 30% of the mice treated with TPA had tumors compared to 100% of the mice treated without PFE. This result was attributed to suppression of inflammation (COX-2, MAPK, NFκB) and ornithine decarboxylase, a tumor growth marker [11].

The effects of flavonoid-rich PJ and FPJ and pomegranate peel extracts on differentiation HL-60 (ability of cancer cells to transform into normal cells) and proliferation of human leukemia cells were investigated. Structural similarities between plant flavonoids and retinoids, the latter of which are established as differentiation-promoting agents, led to the hypothesis that flavonoid-rich pomegranate extracts may have similar effects on differentiation.

In vitro studies confirmed that both FPJ extract and pericarp extract strongly promoted cell differentiation and inhibited proliferation in her HL-60 cell culture. The effect of PJ on cell differentiation was less pronounced. This study suggests another mechanism by which pomegranate components confer their anti-carcinogenic effects [2].

Pomegranate fruit, pomegranate juice, seeds and seed oil, due to their antioxidant and anti-proliferative effects (growth inhibition, cell cycle disruption and apoptosis), can be used to treat prostate cancer, breast cancer, skin cancer, colon cancer, lung cancer and oral cancer. Effective against cancer and leukemia [12].

Dietary pomegranate seed oil significantly reduced the incidence and incidence of azoxymethane-induced colon cancer (measured as number of tumors per rat) [3]. In this experiment, pomegranate seed oil was added to the AIN-76A feed. Increasing concentrations of pomegranate seed oil at 0.01%, 0.1%, and 1% (w/w) did not show a dose-response effect, although anticarcinogenic effects were observed at all doses used in the study.

Pomegranate seed oil and fermented fruit juice polyphenols tend to inhibit breast cancer cell proliferation and invasion and promote breast cancer cell apoptosis [8].

Fermented pomegranate juice polyphenols consistently showed twice the antiproliferative effect compared to fresh pomegranate juice polyphenols. Lung cancer studies have shown that PFE (pomegranate fruit extract) is an effective treatment for lung cancer. This result suggests that PFE can be used as a chemopreventive agent against lung cancer [13].

The effects of pomegranate oil, seed oil, fermented fruit juice polyphenols, and pericarp polyphenols on the proliferation of human prostate cancer cells in vivo have shown that they exhibit significant antitumor activity against human prostate cancer. Pomegranate fruit extract was shown to inhibit cell proliferation and cause apoptosis in highly aggressive human prostate cancer PC-3 cells [1]. The combination of fermented pomegranate juice polyphenols and pericarp polyphenols was also tested for in vitro growth of human prostate cancer cell line DU 145. Super-additive and synergistic effects have been experimentally demonstrated. These studies provide evidence that pomegranate consumption may slow the progression of prostate cancer.

Pomegranates have inhibitory effects on various types of cancer, including prostate cancer, breast cancer, colon cancer, and lung cancer. Caffeic acid phenethyl ester (CAPE), a Pg compound also derived from honey bee propolis, inhibits MMP and TIMP-2 mRNA in HT1080 human fibrosarcoma cells, as detected by reverse transcriptase polymerase chain reaction (RTPCR). Showed a dose-dependent decrease in levels. Gelatin zymography analysis confirmed this compared to controls. This study demonstrates the role of CAPE as an effective anti-metastatic agent that can significantly inhibit the metastatic and invasive ability of malignant cells [14].

Pg shows dose-dependent inhibitory effects on NFkB-dependent reporter gene expression associated with proliferation, invasion, and motility in aggressive breast cancer phenotypes. This effect is the reason for the decrease in RhoC and RhoA protein expression, suggesting that these extracts play a role in reducing the metastatic potential of aggressive breast cancers [7].

Four pure chemicals from Pg fruit, ellagic acid, caffeic acid, luteolin and punicic acid, had a significant effect on the in vitro invasion potential of human PC-3 prostate cancer cells in a test using an engineered Matrigel membrane. It has been proven to be an effective inhibitor. *Punica granatum* extract (PE) inhibits the proliferation of a mouse breast cancer cell line (WA4) derived from MMTV-Wnt-1 mouse mammary tumor by blocking cell cycle progression in G0/G1 over time and concentration.

Pg juice-derived ellagitannins and their metabolites urolithins dose- and time-dependently affect cell proliferation and clonogenic efficiency of HT-29 cells through cell cycle arrest in G0/G1 and G2/M. Apoptosis is induced following each stage of the cell cycle.

Furthermore, Pg treatment induced a dose-dependent arrest in the G0/G1 phase of the cell cycle, which was assessed by DNA cell cycle analysis in lung cancer cell lines (A549) [2]. Found that pretreatment of normal human epidermal keratinocytes (NHEK) with µg increased UVA-induced cell cycle arrest in the G1 phase of the cell cycle.

Furthermore, the androgen-independent cell lines DU 145 showed a significant increase from 11% to 22% in G2/M cells when treated with (35 µg/ml) cold-pressed Pg oil ($p < 0.05$).

Ellagic acid, a phenolic compound probably belonging to Pg, inhibits T24 human bladder cancer in vitro through induction of G0/G1 phase arrest, increase of p53 and p21, and decrease of cyclin-dependent kinase (Cdk2). Induces cell cycle arrest and apoptosis in cells. Cycle progression at the G1-S phase restriction point and is inhibited by the tumor suppressor p16 (INK4a).

It has been reported that the N terminus of some truncated p16 molecules (INK4a) is not important for interaction with Cdk4. However, the C-terminal domain of p16 (INK4a) has previously been shown to be sufficient to cause cell cycle arrest, growth inhibition, and CDK4/6 interaction [13,14].

Pg extract and punicic acid, a long-chain omega-5 polyunsaturated fatty acid derived from Pg, inhibited apoptosis in both estrogen-sensitive breast cancer cell lines (MDA-MB-231) and estrogen-sensitive breast cancer cell lines has been shown to induce A cell line derived from MDA-MB-231 cells (MDA-ERalpha7) through lipid peroxidation and PKC (protein kinase C) signaling pathways. It also causes destruction of the cell's mitochondrial membranes [15].

We investigated the relationship between pg-induced apoptosis and the IGF/IGFBP system in human prostate cancer cells (LAPC4). POMx (a potent Pg extract prepared from peel and seedless seeds) and IGFBP-3 have been shown to synergistically stimulate apoptosis. Inhibition of cell proliferation increased JNK phosphorylation and decreased Akt and mTOR activation.

Pg treatment of normal human epidermal keratinocytes (NHEKs) induces UVB-mediated activation of MAPK and NF-κB signaling pathways, as well as other signal transducers including transcription 3 (STAT3), PKB/AKT, ERK1/2 and Inhibited activators of apoptotic signaling pathway, mTOR, PI3K, Bcl-X(L) (anti-apoptotic protein), Bax and Bad (pro-apoptotic protein).

Conclusion:

Research in the field of cancer treatment is vast and requires significant investment. Many researchers struggle to push the boundaries of clinical trials. Much of the research remains at the laboratory scale or pharmacological level. The medicinal properties of the *Punica granatum* plant have been well utilized by our ancestors for centuries. This provides additional benefits to the bioactive molecules of this plant. To interpret and discover the potential of *Punica granatum* as a reliable anticancer therapeutic agent, further analysis of the bioactive components of *Punica granatum* is required.

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Conflict of interest:

None to be declared.

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